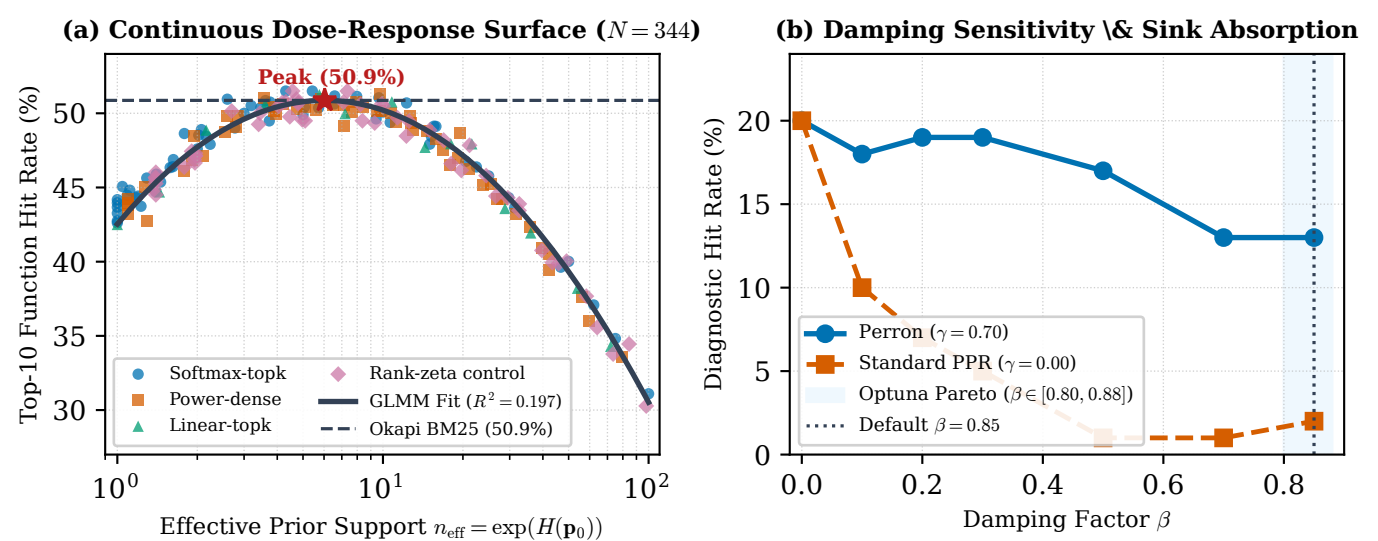
(a) Continuous Dose-Response Surface ($N = 344$)
Peak (50.9%)
Top-10 Function Hit Rate (%)
Effective Prior Support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$
Softmax-topk
Power-dense
Linear-topk
Rank-zeta control
GLMM Fit ($R^2 = 0.197$)
Okapi BM25 (50.9%)
(b) Damping Sensitivity \& Sink Absorption
Diagnostic Hit Rate (%)
Damping Factor β
Perron ($\gamma = 0.70$)
Standard PPR ($\gamma = 0.00$)
Optuna Pareto ($\beta \in [0.80, 0.88]$)
Default $\beta = 0.85$